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DHRUV SHARMA

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EDUCATION

University of California, Berkeley — **Berkeley, CA**
B.A. in Applied Mathematics — *Expected Graduation: May 2030*

BASIS Independent Silicon Valley — **San Jose, CA**
High School Diploma with High Honors (GPA: 4.612) — *August 2019 – May 2026*

Stanford University: Summer Session — **Stanford, CA**
Biotechnology & Symbolic Systems (GPA: 4.171) — *June 2025 – August 2025*

RESEARCH PROJECTS

QuMOS: Using Quantum Computing for Accelerated Multi-Omics Analysis in Vaccine Development

IEEE-EMBS Presenter | JSHS Regional Semifinalist | Synopsys Category & Special Judge Award | UW Faculty Review

- This research explores quantum computing in multi-omics analysis, proposing quantum algorithms using Grover's Algorithm and Quantum Feature Maps for efficiently processing genomic datasets like the SARS-CoV-2 spike protein
- The research outperformed traditional machine learning models, improving mutation analysis and accelerating the analysis of large multi-omics datasets for faster and more accurate biological insights.

Machine Learning Driven Discovery of Novel EEG Biomarkers in SYNGAP1-Associated Genetic Epilepsy

Research Internship at Stanford University — Submitting to ACNS Journal of Clinical Electrophysiology

- This research project, conducted at the Dr. Lee-Messer Lab, aims to discover novel EEG biomarkers in genetic epilepsies using machine learning to extract oscillatory patterns and high-frequency oscillations (HFOs) linked to seizure networks
- These are analyzed through machine learning tools such as automatic HFO detection, pretrained EEG neural networks, and deep learning models for spike classification to identify which features most reliably distinguish SYNGAP1 from healthy brain activity.

INTERNSHIPS AND EXPERIENCES

Cards 4 Kids (Stanford University) — *January 2025 – Present*
Director of Partnerships & Programs

- Only high schooler among Stanford undergrads leading program coordination, volunteer training, and impact tracking across 20+ units at Lucile Packard Children's Hospital and Ronald McDonald House
- Expanded chapters through outreach to 50+ pre-med and health college clubs; recruited D1 college athletes for card-giving events and ran sports-themed activities with pediatric patients
- Created custom trading-card templates signed by athletes from Stanford, San Jose State, and UC Santa Cruz, coordinating logistics and athlete recruitment for the Stanford chapter

Lee-Messer Lab, Stanford Department of Pediatric Neurology — *March 2025 – Present*
Research Intern

- Applied computational analysis and EEG data interpretation to explore seizure patterns and inform treatment strategies
- Presented research at the Stanford Knight Initiative for Brain Resilience; submitting paper to an ACNS journal

Flowers4Smiles — *November 2024 – Present*
Founder & President

- Founded a community initiative delivering handmade bouquets and cards to seniors and caregivers across 20+ Bay Area hospitals and care centers; building florist partnerships toward registered nonprofit status
- Built partnerships with local florists and volunteers while developing plans to formalize the program into a registered nonprofit to expand regional outreach in the future.

Selected Awards & Publications

3× IEEE-EMBS Conference Publications (NER, BHI, BSN – 2025) • JSHS Semifinalist (2025) • 2× Synopsys Science Fair Winner, Computational Biology (2024, 2025) • Presenter, Wu Tsai Neuroscience Symposium & Stanford Knight Initiative

SKILLS AND INTERESTS

Programming: Python, Qiskit, MATLAB, R, PyTorch, TensorFlow, Keras, Scikit-Learn, EEG Signal Processing
Interests: Tennis, Running, Basketball, Football, Traveling, Backpacking, Exploring Coffee Shops, Reading, Cooking, Neuroscience Articles, Producing Music (DJ), History